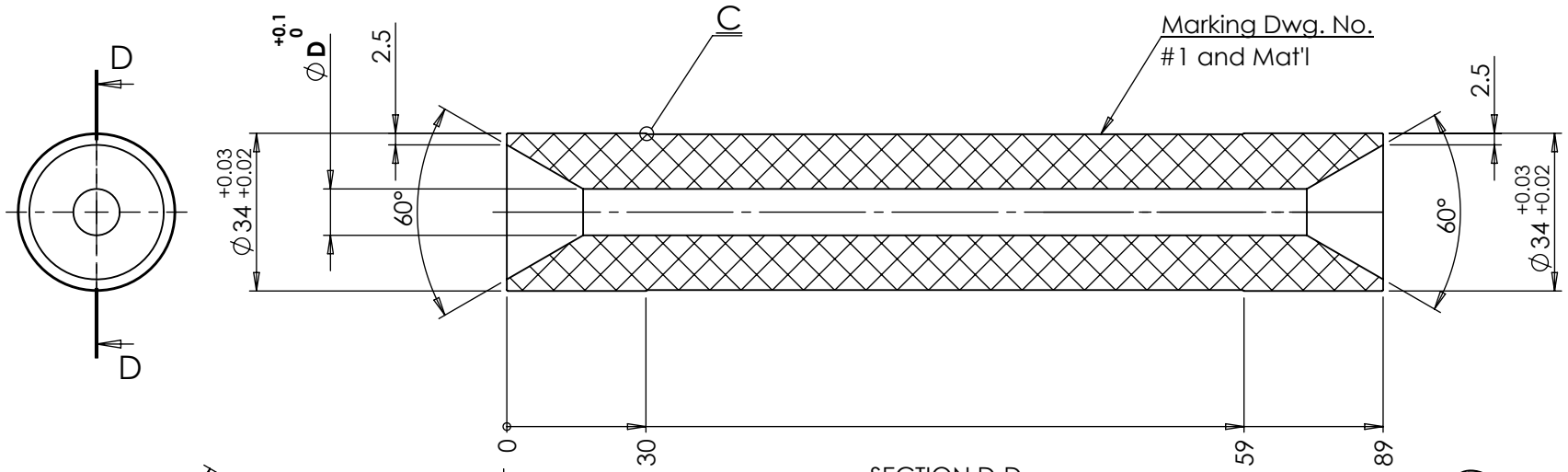
PART No. TGM-12-390-2A-xx

REV. 7

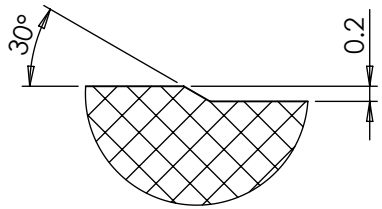
Rt=12

☑	Mat'l Tick by user
☐	Pom black
☐	PP Gray
☐	Peek Gray
☐	Pom white

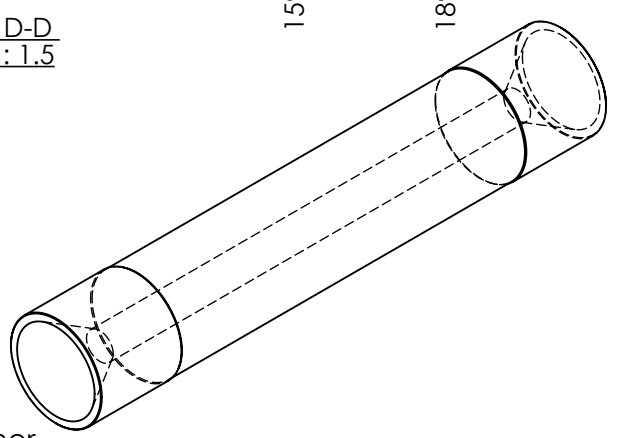
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SECTION D-D
SCALE 1 : 1.5



DETAIL C
SCALE 10 : 1



Use for: Pusher

Note:
1. Break all Corner approx by C0.2
2. General Tolerance ±0.1
3. Vendor Ass'y TGM-12-390-1A = TGM-12-390-2A (Both Side)

Item	D +0.1 0	For Mat'l	Check
-01	ϕ10	ϕ8.51 to ϕ9.5	RQ.346/16
-02	ϕ12	ϕ10.51 to ϕ11.5	RQ.346/16
-03	ϕ7	ϕ5.51 to ϕ6.5	RQ.346/16
-04	ϕ8	ϕ6.51 to ϕ7.5	RQ.2875
-05	ϕ18	ϕ16.9 to 17.8	For PMA.
-06	ϕ9	ϕ7.51 to ϕ8.5	RQ.2969
-07	ϕ6	ϕ5.01 to ϕ5.5	RQ.118/14
-08	ϕ6.5	ϕ6.1 to ϕ6.2	RQ.773/14
-09	ϕ4.9	ϕ4.4	Preecha RQ.304/26
-10			
-11			
-12			
-13			
-14			
15			

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE

UNLESS SPECIFIED C0.1 OR R0.1
GENERAL TOLERANCE
JISB0405 MACHING WORKING

LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM-16 | Q'ty 1

MATERIAL 4 See table

FINISH -

HEAT TREATMENT -

List Request No. 117/15

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	5 CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 9 Mar' 11

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Bush (A)

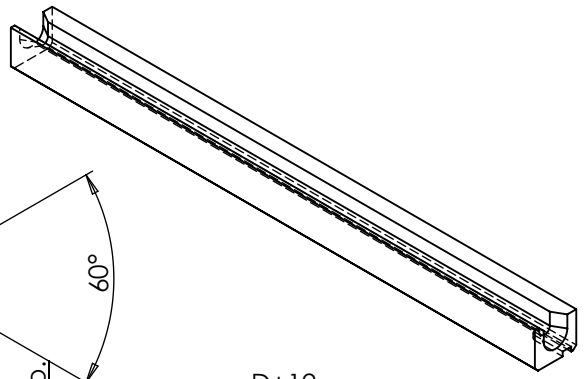
PART No. TGM-12-390-2A-xx

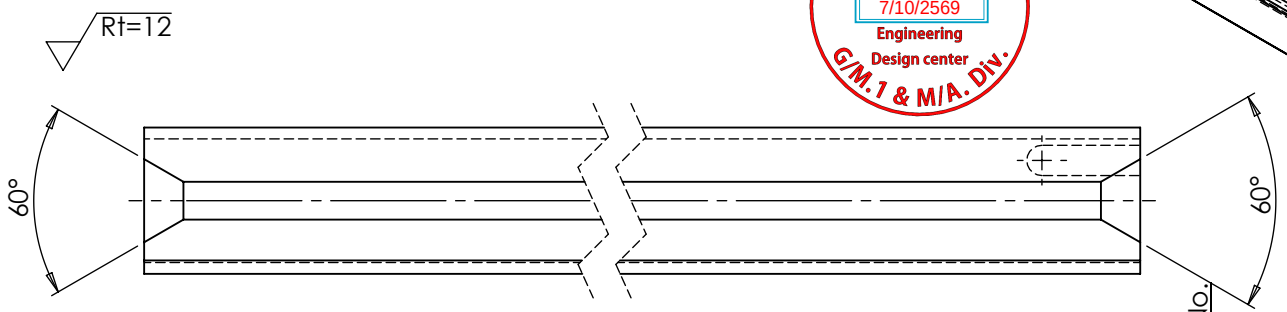
REV. 1

PART No. TGM-12-390-3A-xx

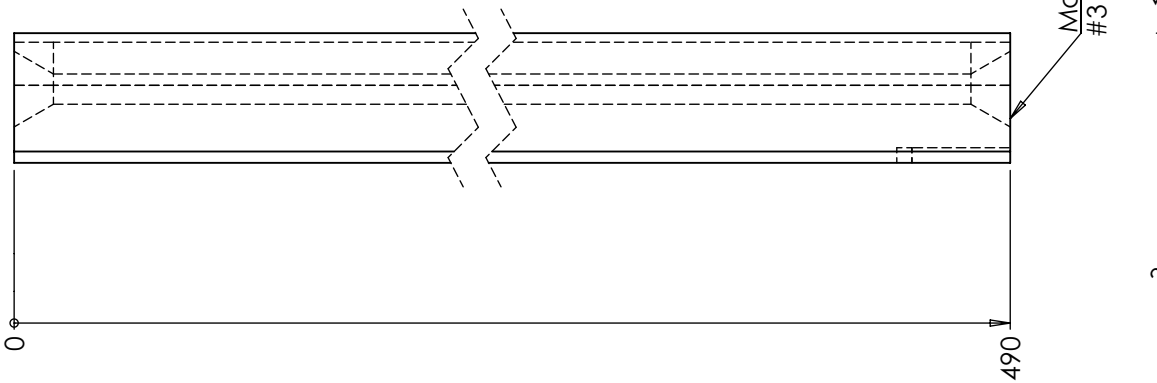
3 REV.

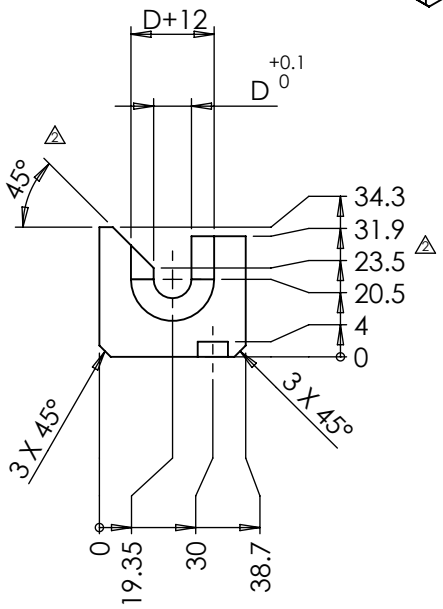
MinebeaMitsumi Inc.
Copy issued Uncontrolled
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G/M.1 & M/A. Div.

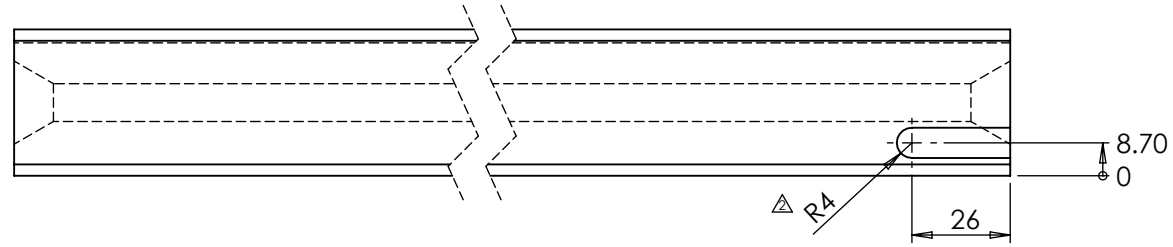




Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1







Item	D +0.1 0	For Mat'l	Check
-01	∅13	∅11	Preecha
-02	∅11.2	∅9.2	Preecha
-03	∅7	∅5.12	Preecha
-04	∅11	∅9	Preecha
-05	∅8	∅6.51	Preecha RQ.2875
-06	∅18	∅16	Preecha For PMA.
-07	∅9	∅7.7	Preecha RQ.2969
-08	∅6	∅5.12	Preecha RQ.118/14
-09	∅15	∅14	Preecha RQ.263/16
-10	∅13	∅11.6 to 12.8	Preecha RQ.271/16
-11	∅4.9	∅4.4	Preecha RQ.304/26
-12			
-13			
-14			
-15			

☑	Mat'l Tick by user
☐	Pom black
☐	PP Gray
☐	Peek Gray

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	21 Oct'13	Preecha	Add: 23.5, Delete 4 etc.; Rq. 3276
3	----	22 Jul'16	Preecha	Add: 30 etc.;

UNLESS SPECIFIED C0.1 OR R0.1
GENERAL TOLERANCE
JISB0405 MACHING WORKING

LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. BM-16 | Q'ty 5
MATERIAL See table
FINISH -
HEAT TREATMENT -

List Request No. 3081
3 RD ANGLE
SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 21 Mar' 11

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: G-Channel (A)

PART No. TGM-12-390-3A-xx

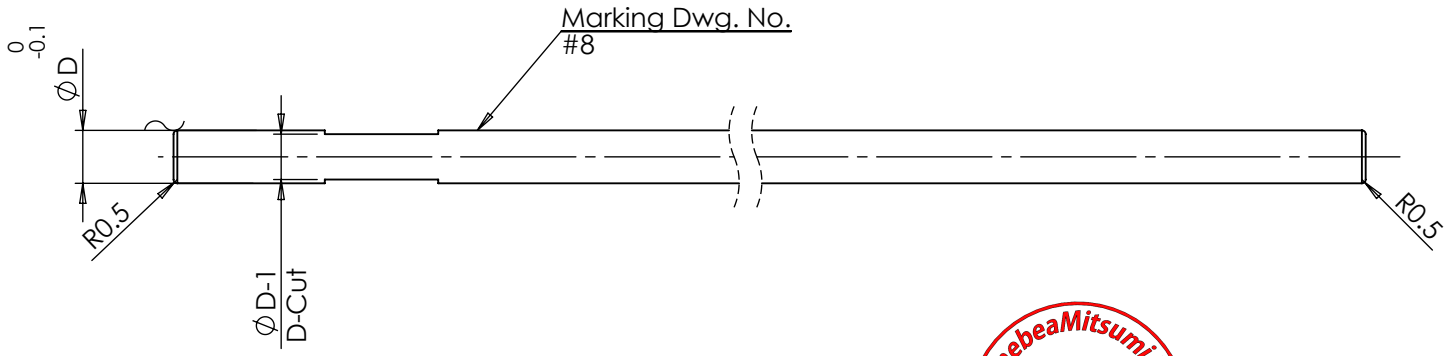
3 REV.

PART No. TGM-12-390-12B-xx

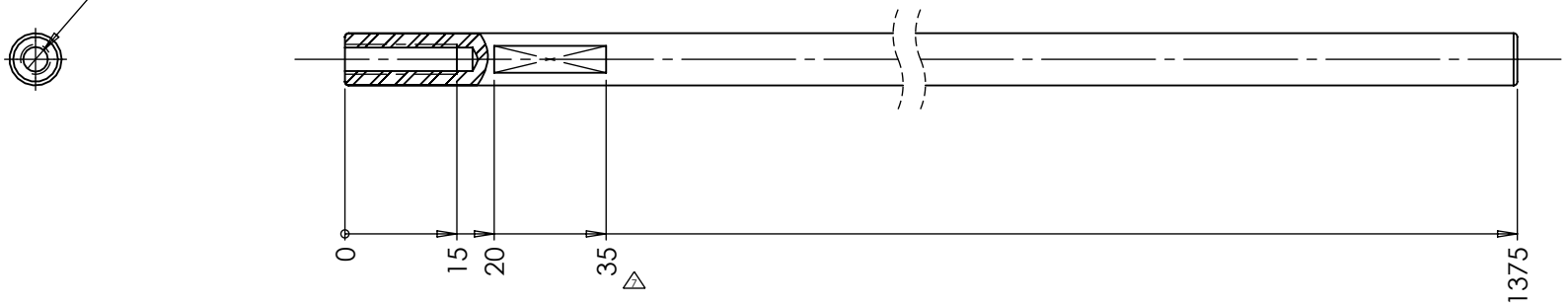
8 REV.

△3 Formula: ∅D = OD. Mat'l - 0.5

Rt=12 (~)



△3△2 Screw = M
Left Hand
See table



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G/M.1 & M/A. Div.

Item	∅D	△3 M Screw Left	Mat'l	Check
-16	△8 12.7	M8x1.25	△3 13.2	Preecha
-17	16.5	M8x1.25	17	RQ.705/16
-18	19.5	M8x1.25	20	RQ.121/17
-19	17.5	M8x1.25	18	RQ.13/12
-20	4	M3x0.5	4.4	Preecha RQ.304/26
-21				
-22				
-23				
-24				
-25				
-26				
-27				
-28				
-29				
-30				

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

△3 Use for: Model Pusher

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
8	----	9 Mar'20	Preecha	Change: Item -13 to-16 Change spec. gap = 0.5
3	----	9 Oct'14	Preecha	Change: Item -05 16 ==> 15.5 etc.; Rq. 601/14
4	----	1 Sep'16	Preecha	Change: M5x0.8 ==> M6x1, M8x1.25 etc.;
5	----	20 Oct'16	Preecha	Change: M5x0.8 ==> M6x1, M8x1.25 etc.; Item -01, -14
6	----	20 Dec'16	Preecha	Change: R1 ==> R0.5 Rq. 646/16
7	----	2 Dec'17	Preecha	Change: 25 ==> 35 Rq. 602/17

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM-165 | Q'ty 1

MATERIAL SUS 304

FINISH -

HEAT TREATMENT -

List Request No. 2938

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chonknok	Preecha	Preecha	Preecha

DATE 25 May' 13

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Shaft Pusher Mat'l (B)

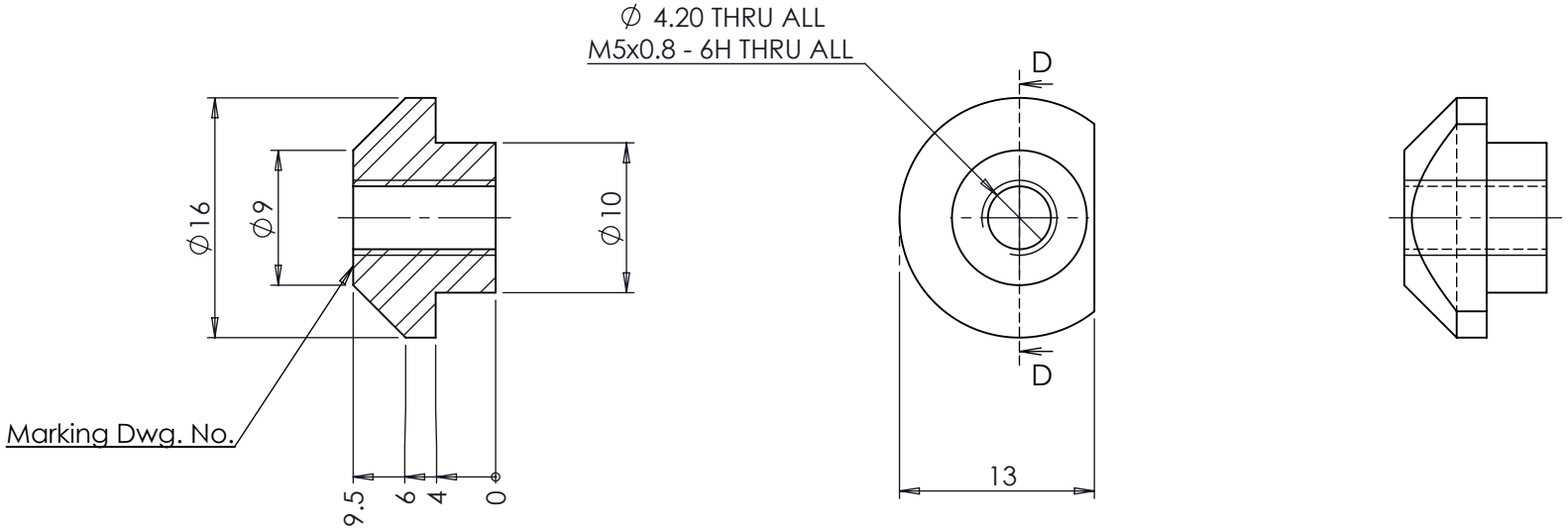
PART No. TGM-12-390-12B-xx

8 REV.

PART No.
TGM-12-390-13

1 REV.

Rt=6



SECTION D-D
SCALE 2 : 1

Note: 1. Break all Corner approx by C0.1
2. General Tolerance ±0.1

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Engineering Design center
G/M.1 & M/A. Div.

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. BM-165 | Q'ty 1
MATERIAL SUS 304
FINISH -
HEAT TREATMENT -

List Request No. -
3 RD ANGLE
SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chonknok	Preecha	Preecha	Preecha

DATE 20 Oct' 14

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Nut lock plate

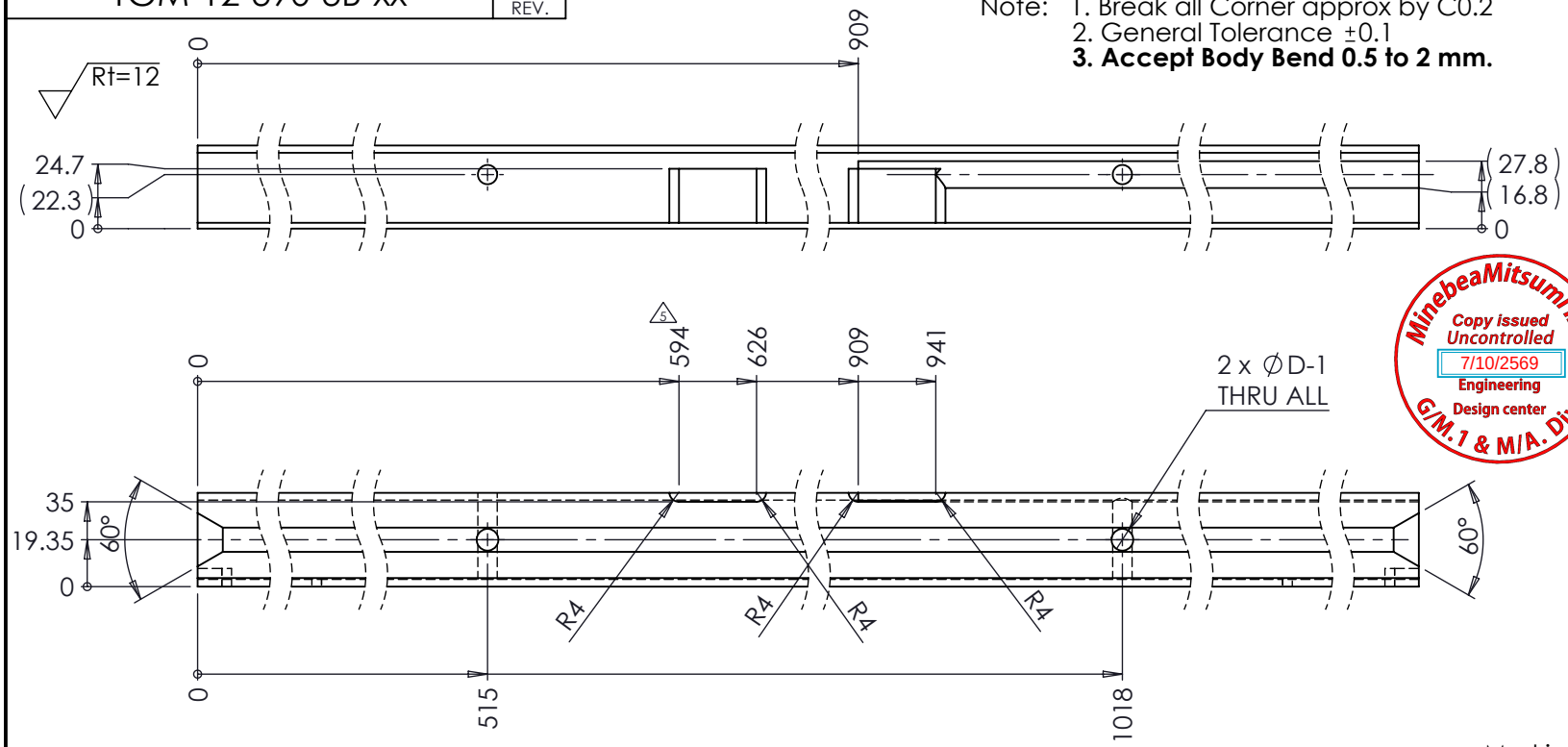
PART No. TGM-12-390-13

1 REV.

PART No.
TGM-12-390-3B-xx

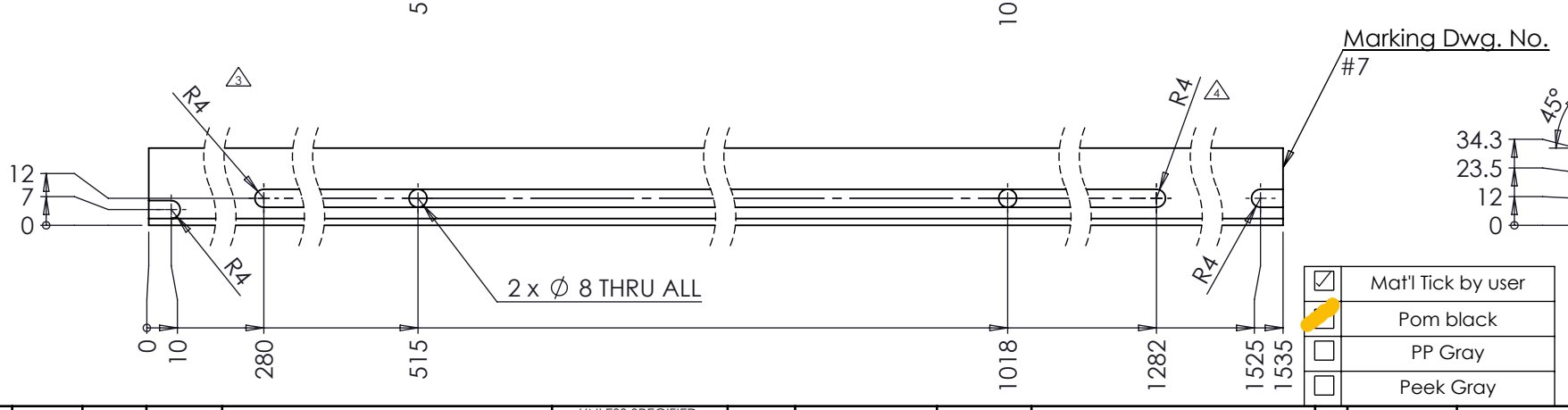
7 REV.

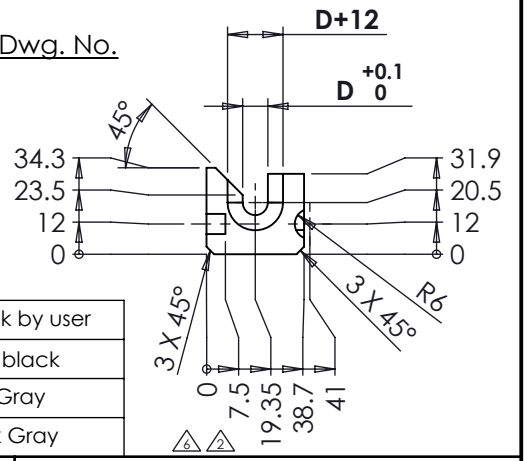
Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1
3. Accept Body Bend 0.5 to 2 mm.



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Item	∅D	for Mat'l	Check
-01	∅10	∅8.51 to ∅9.5	Preecha
-02	∅12	∅10.5 to ∅11.5	Preecha
-03	∅7	∅5.51 to ∅6.5	Preecha
-04	∅11	∅9.51 to ∅10.5	Preecha
-05	∅8	∅6.51 to ∅7.5	Preecha
-06	∅18	∅16	Preecha
-07	∅9	∅7.51 to ∅8.5	Preecha
-08	∅6	∅5.01 to ∅5.5	Preecha
-09	∅6.5	∅6.1 to ∅6.2	Preecha
-10	∅4.9 (7)	∅4.4	Preecha RQ.304/26
-11			
-12			
-13			
-14			
-15			





☑	Mat'l Tick by user
☐	Pom black
☐	PP Gray
☐	Peek Gray

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	24 Aug'15	Preecha	Change : 2.5 ==>7.5 etc;.
3	----	11 Jan'16	Preecha	Add : Slot etc,. RQ.22/16
4	----	23 Feb' 16	Preecha	Change : R4 ==> R4 ▽3 etc;.
5	----	31 Mar'16	Preecha	Add : Oil Key etc;. Rq : 142/16
6	----	7 Oct'26	Preecha	Change : 19.4 ==> 19.35 etc;. RQ.304/26
7	----	7 Oct'26	Preecha	Change : ∅4.4 ==> ∅4.9 RQ.304/26

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM16 | Q'ty 1

MATERIAL See table

FINISH -

HEAT TREATMENT -

List Request No. 317/15

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 8 Jul' 15

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

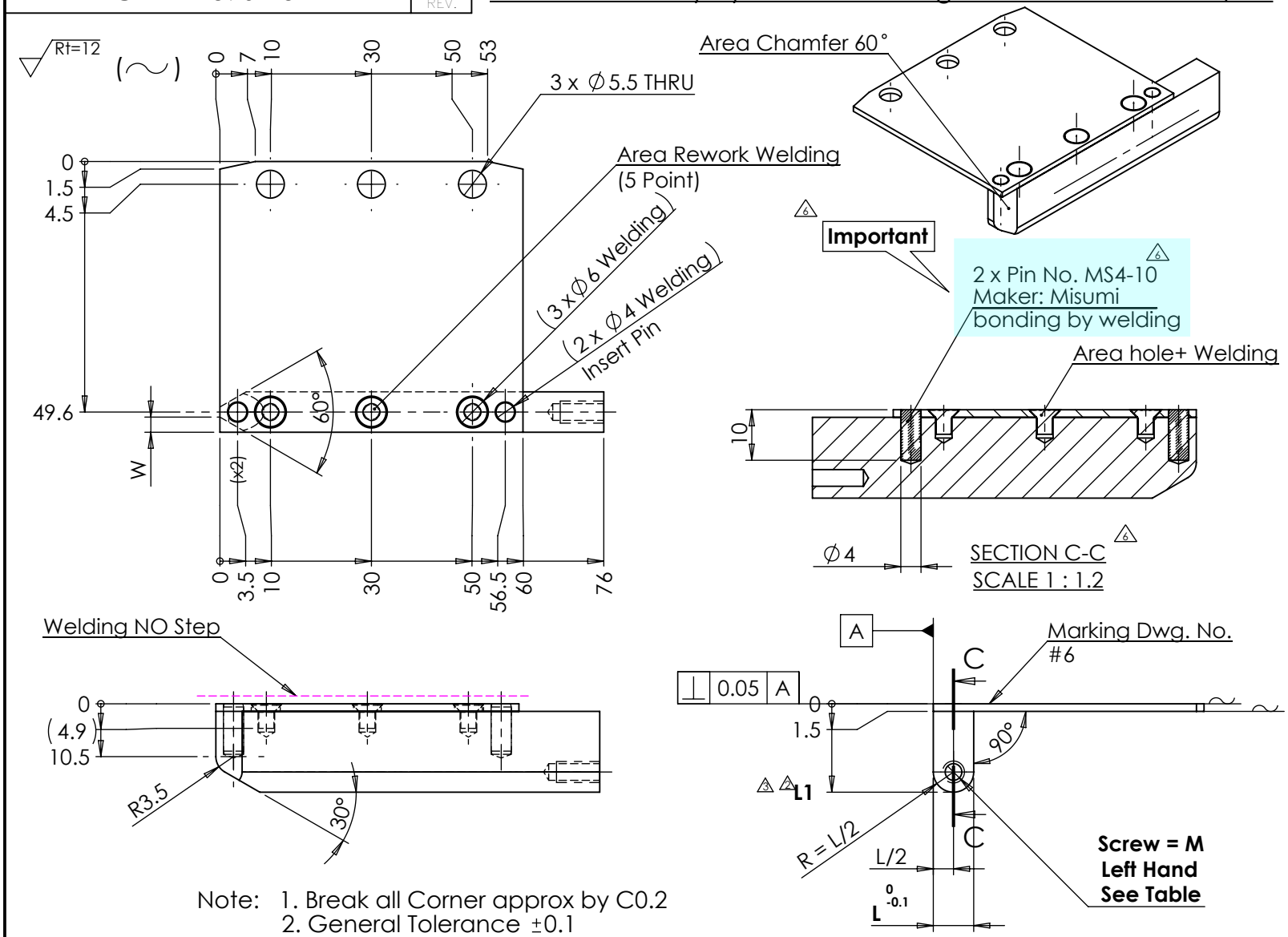
TITLE: G-Channel (BM-16)

PART No. TGM-12-390-3B-xx

7 REV.

PART No. TGM-12-390-25-xx | 6 REV.

For: Vendor Ass'y by Process Welding Close Hole Send to G/M1



Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Item	L	L1	W	M Screw Left	Use For Mat'l	Check
-16	17.5	22	6	M8x1.25	∅18	Preecha
-17	4	15.8	2	M3x0.5	∅4.4	Preecha RQ.304/26
-18						
-19						
-20						
-21						
-22						
-23						
-24						
-25						
-26						
-27						
-28						
-29						
-30						

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G/M.1 & M/A. Div.

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	----	ENGINEERING RELEASE
2	----	16 Jul'16	Preecha	Change: 17.5 ===> 21.5 Rq. 346/16
3	----	31 Aug'16	Preecha	Change: M5x0.8 ===> M8x1.25 etc.: Rq. 346/16
4	----	20 Oct'16	Preecha	Change: M5x0.8 ===> M8x1.25 etc.: Rq. 346/16
5	----	16 Jun'17	Preecha	Change: M5x0.8 ==> M8x1.25 etc.; Item -08, -11
6	----	3 Feb'26	Preecha	Add: Section etc.; RQ.37/26

UNLESS SPECIFIED C0.1 OR R0.1
GENERAL TOLERANCE
JISB0405 MACHING WORKING

LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S | 6.3S | 25S | 100S | ~ UNCUT

M/C. BM-165 | Q'ty 1
MATERIAL SUS 304
FINISH -
HEAT TREATMENT -

List Request No. 60/15
3 RD ANGLE
SCALE : | SHEET 1/2 | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 13 Feb' 15

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

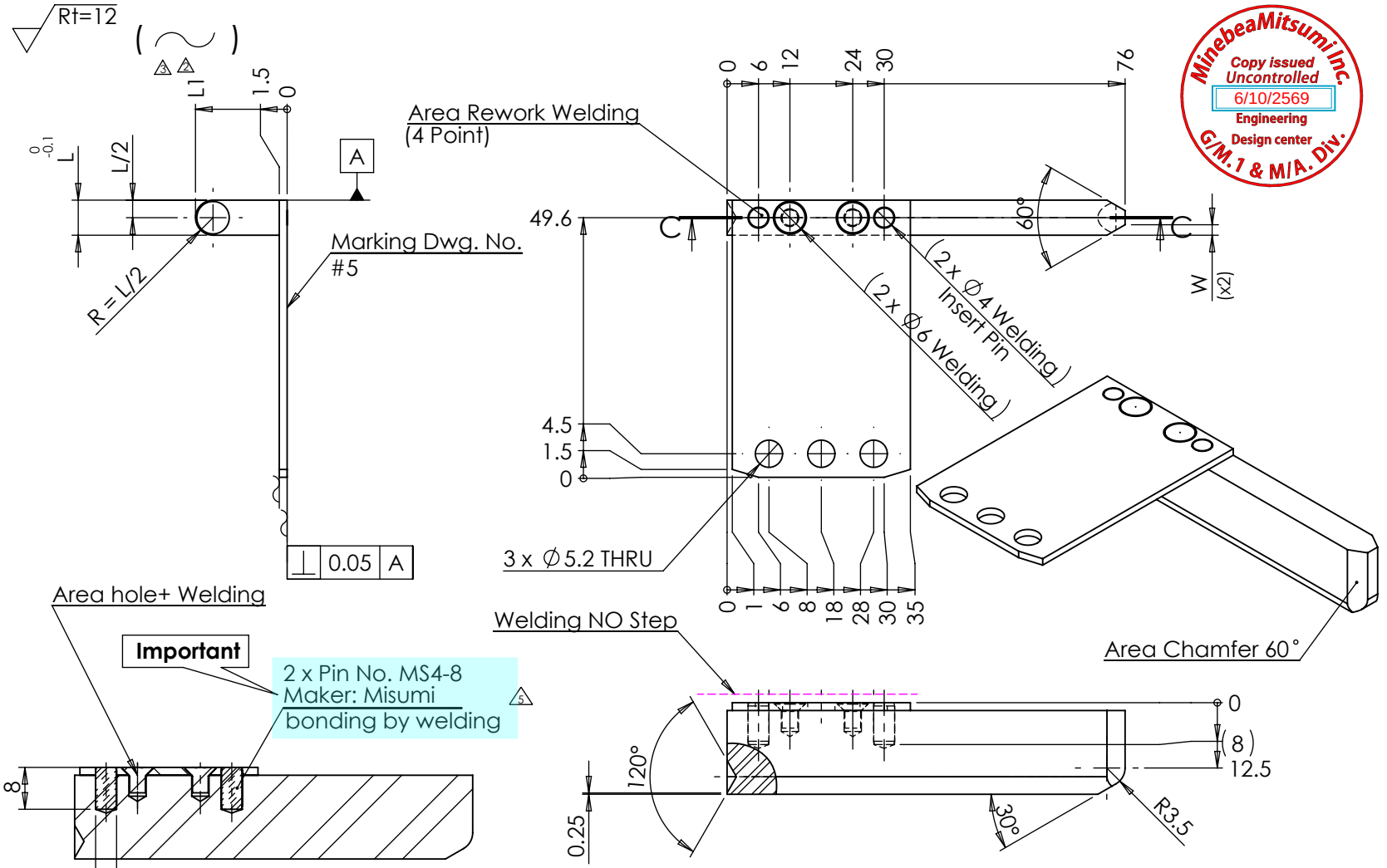
TITLE: Plate Pusher Mat'l (2) 6

PART No. TGM-12-390-25-xx | 6 REV.

PART No. TGM-12-390-26-xx

5 REV.

For: Vendor Ass'y by Process Welding Close Hole Send to G/M1 △4



SECTION C-C △5
SCALE 1 : 1.2

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Item	L	L1	W	Use For Mat'l	Check
-16	17.5	22	6	Ø18	Preecha
-17	4	15.8	2	Ø4.4	Preecha RQ.304/26
-18					
-19					
-20					
-21					
-22					
-23					
-24					
-25					
-26					
-27					
-28					
-29					
-30					

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REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	16 Jul'16	Preecha	Change: 17.5 ===> 21.5 Rq. 346/16
3	----	31 Aug'16	Preecha	Change: 21.5 ===> L1 etc.: Rq. 346/16
4	----	16 Jun'17	Preecha	Cancle Item Screw7 4
5	----	3 Feb'26	Preecha	Add: Section etc.; RQ.37/26

UNLESS SPECIFIED C0.1 OR R0.1
GENERAL TOLERANCE
JISB0405 MACHING WORKING

LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. BM-165 | Q'ty 1
MATERIAL SUS 304
FINISH -
HEAT TREATMENT -

List Request No. 60/15
3 RD ANGLE
SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 13 Feb' 15

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Plate Pusher Mat'l (1)

PART No. TGM-12-390-26-xx

5 REV.